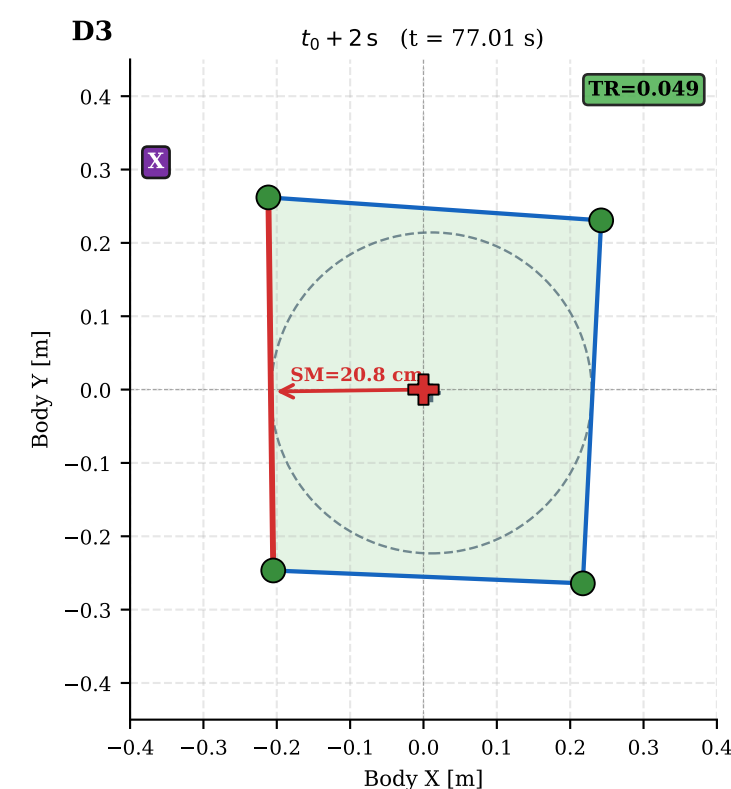
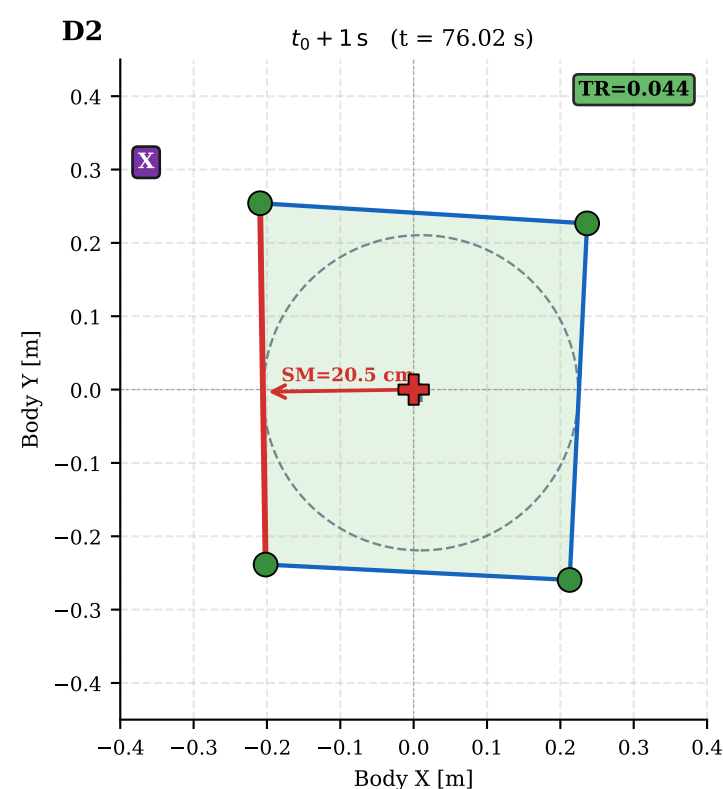
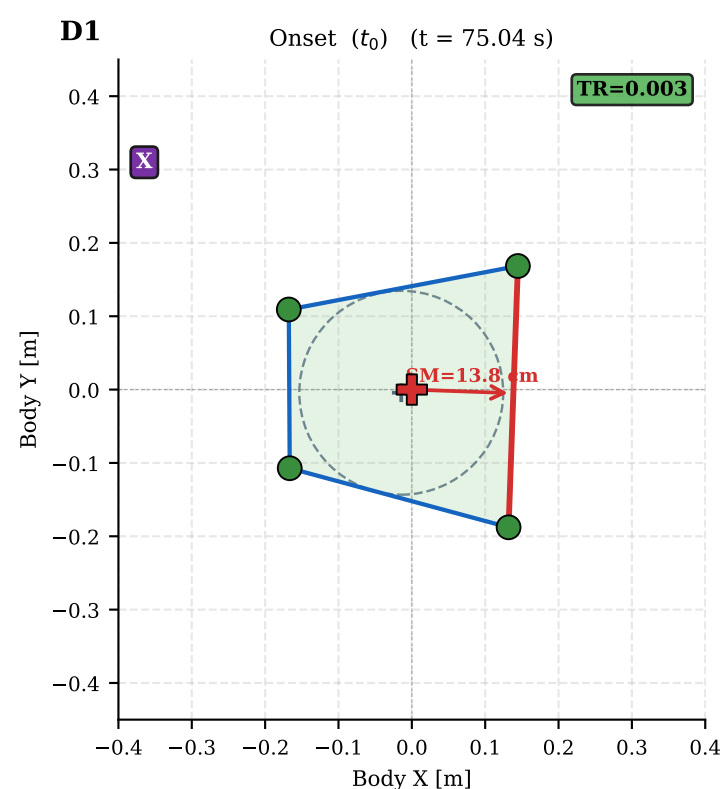
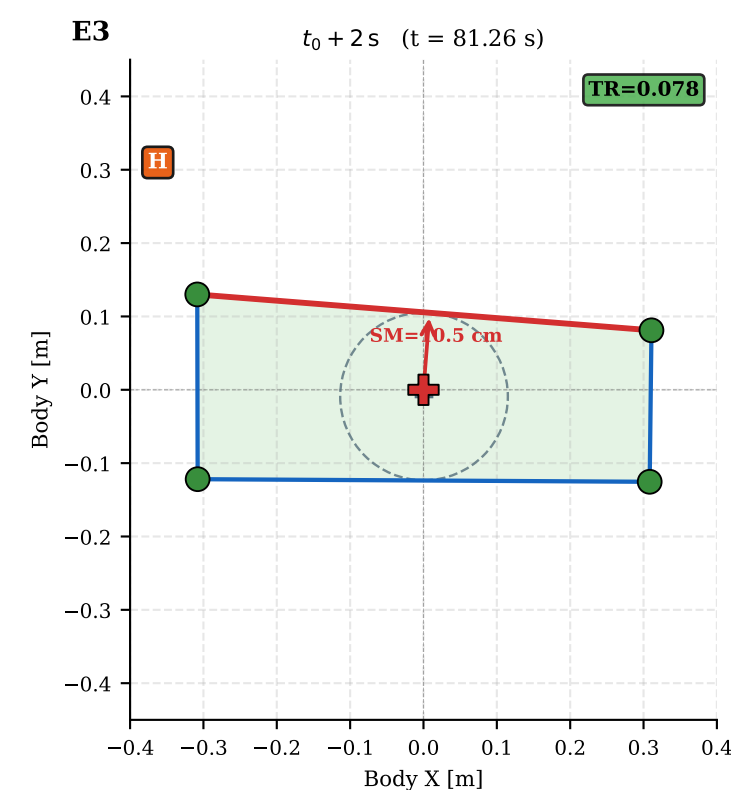
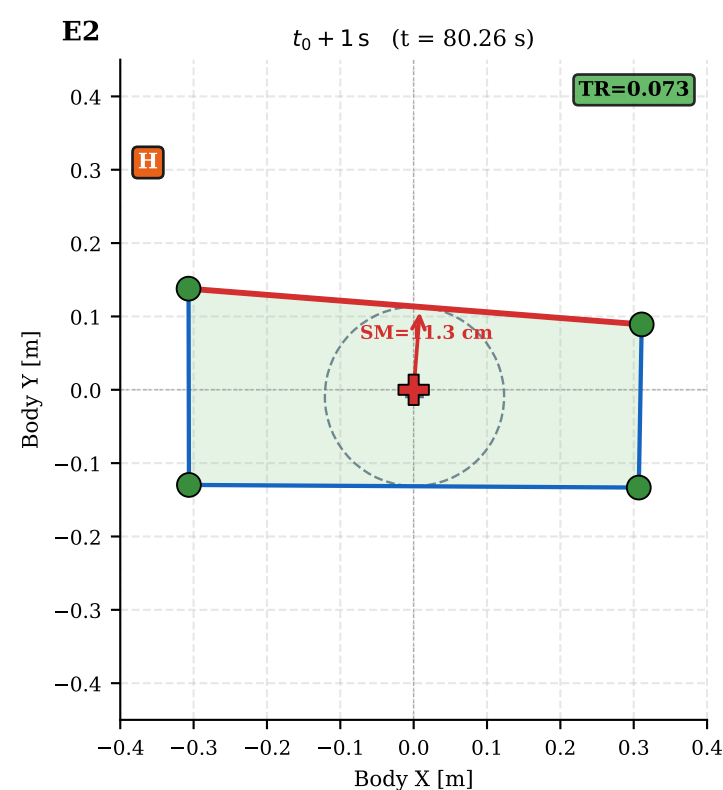
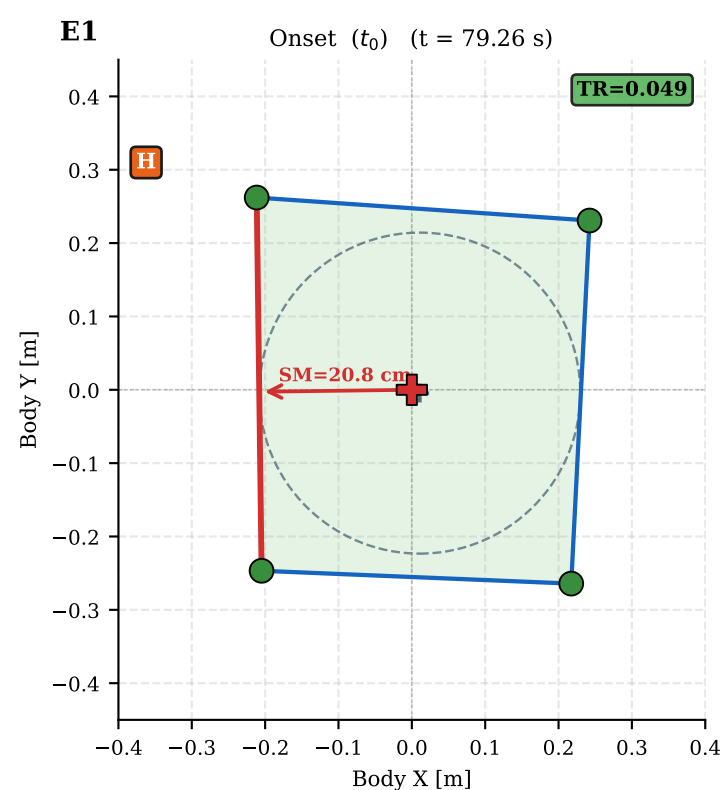


Static Stability Analysis across Locomotion Mode Transitions Generalised McGhee-Frank (1968) Support Polygon — Part 2 of 2

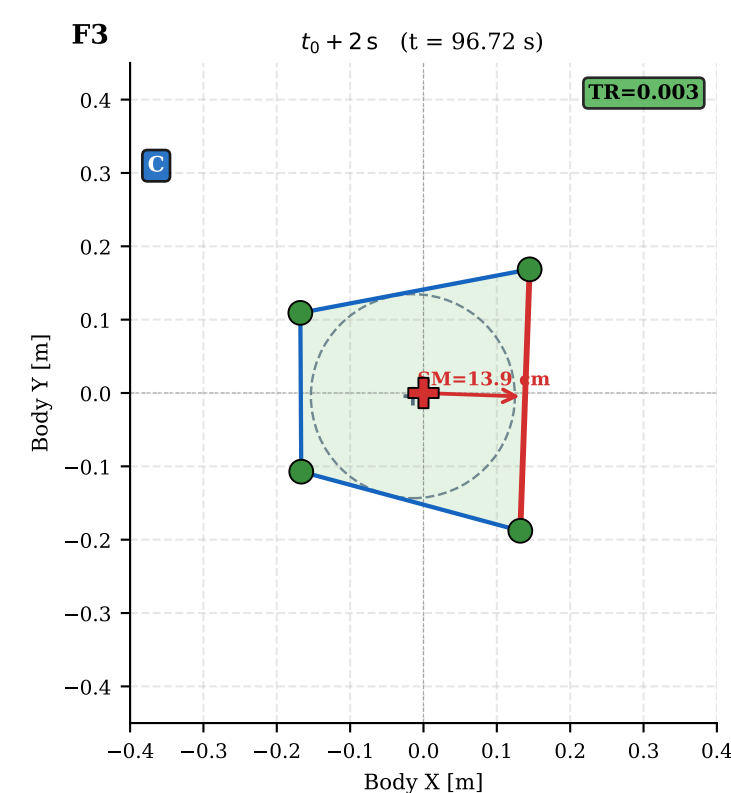
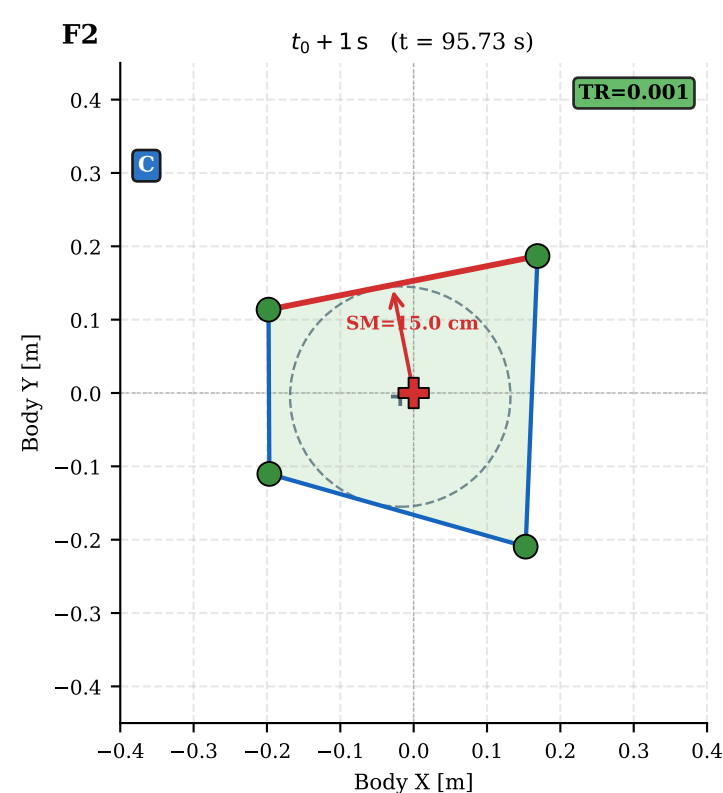
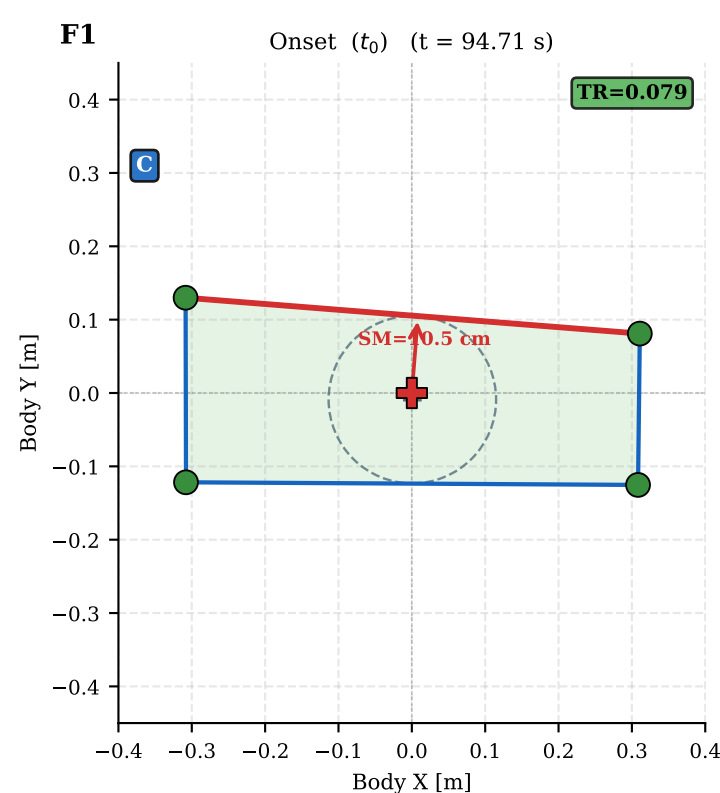
Quadrupedal Walking (C)
↓
Omnidirectional Motion (X)



Omnidirectional Motion (X)
↓
Differential Rolling (H)



Differential Rolling (H)
↓
Quadrupedal Walking (C)



— Support Polygon → SM Arrow (CoM → Critical Edge) ✕ Swing Foot — H — Differential Rolling
— Critical Edge + Polygon Centre + CoM (base_link) — X — Omnidirectional Motion
--- Inradius Circle (r_{in}) ● Stance Feet — C — Quadrupedal Walking